



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



Machine Learning - 216H03C601

Dr. Nirmala Balloorkar

Assistant Professor

Department of Computer Engineering KJSSE

Profile

☐ Educational Qualifications

- Ph.D. in Computer Science and Engineering

☐ Academic and Professional Experience

- Over 16 years of experience in academia and the engineering education domain
- Mentored numerous students at undergraduate and postgraduate levels

☐ Areas of Specialization

- Operating Systems , User Experience Design (UxD), Data Science
- Soft Computing , Machine Learning, Artificial Intelligence
- Full Stack Development (MERN, Django, Flask)

☐ Certifications and Recognitions

- Achieved Elite NPTEL Online Certification in Machine Learning (2019) for academic excellence
- Delivered expert sessions in AICTE–ISTE approved Faculty Development Programs (FDPs) and Short-Term Training Programs (STTPs)
- Served as an evaluator in national-level hackathons

☐ Research Contributions

- Authored 15+ research papers in Scopus-indexed international journals and conferences
- Acted as a reviewer for reputed journals indexed in SCOPUS, Springer, and Elsevier

☐ Academic Roles and Memberships

- Open Elective Coordinator, Faculty Documentation etc.
- Active member of the Indian Society for Technical Education (ISTE)

Course Objectives

- Understand core Machine Learning concepts
- Apply supervised learning to real problems
- Explore data using unsupervised learning
- Get introduced to deep learning and neural networks
- Learn basics of reinforcement learning

Course Structure

The course is divided into **5 modules**:

- Machine Learning fundamentals
- Supervised Learning
- Unsupervised Learning
- Introduction to Deep Learning
- Reinforcement Learning and applications

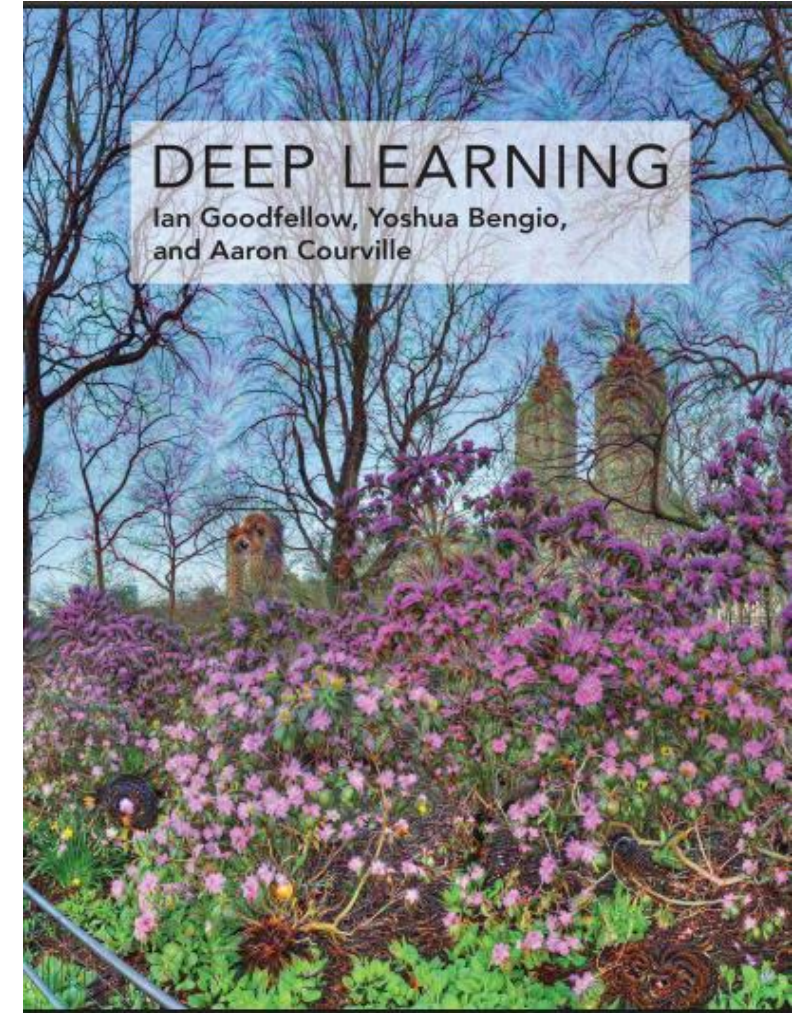
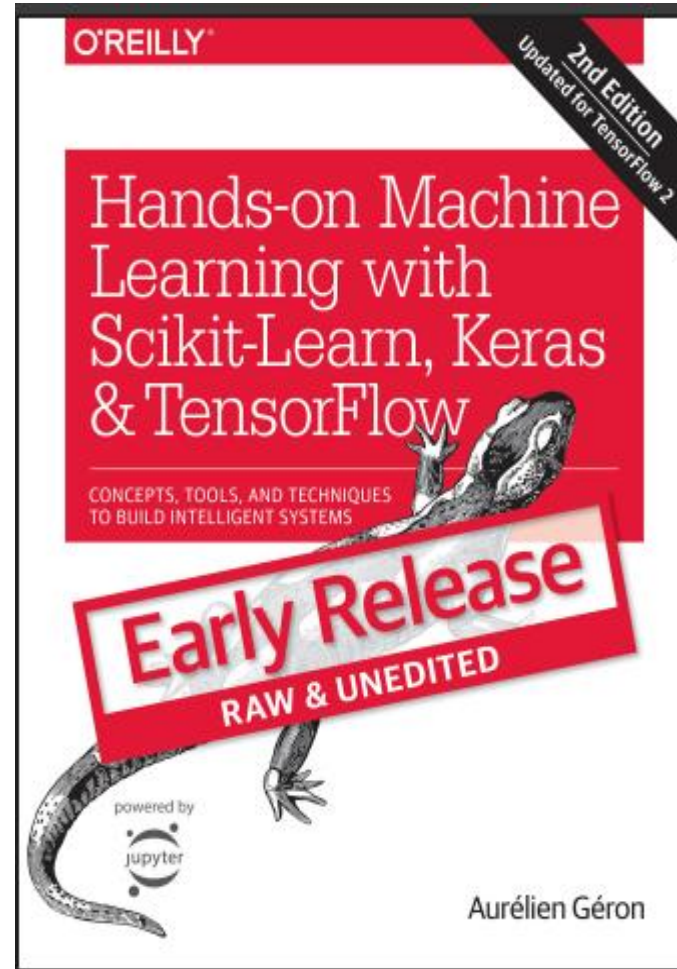
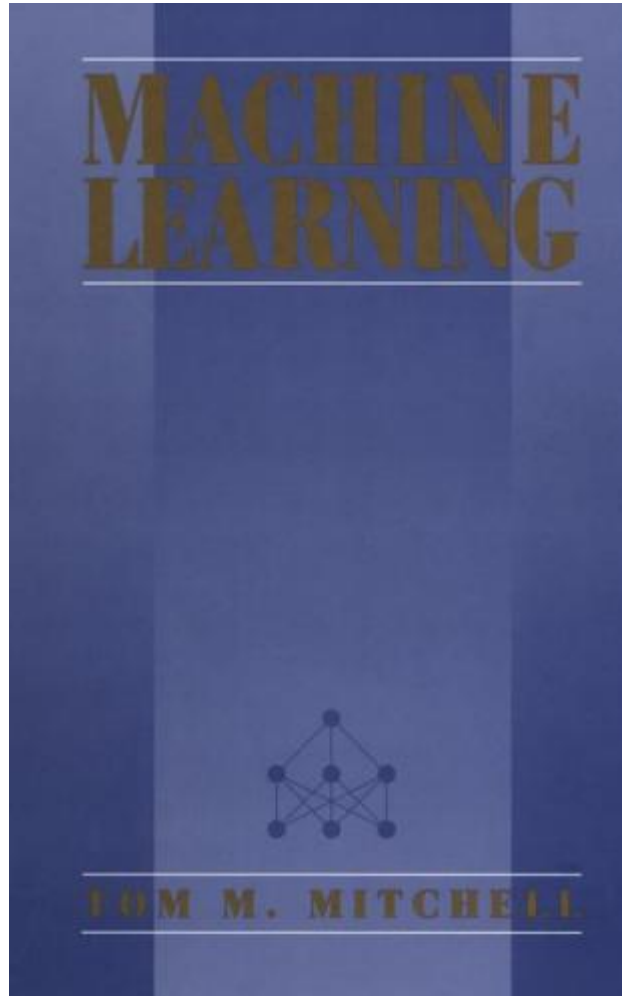
Assessment

Theory :

- Internal Assessment (IA): 20 marks
- In-Semester Evaluation (ISE): 30 marks
- End-Semester Exam (ESE): 50 marks

Total: 100 marks

References



Applications



Disease Diagnosis &
Risk Prediction



Medical Imaging
Analysis



Machine Learning
in Healthcare

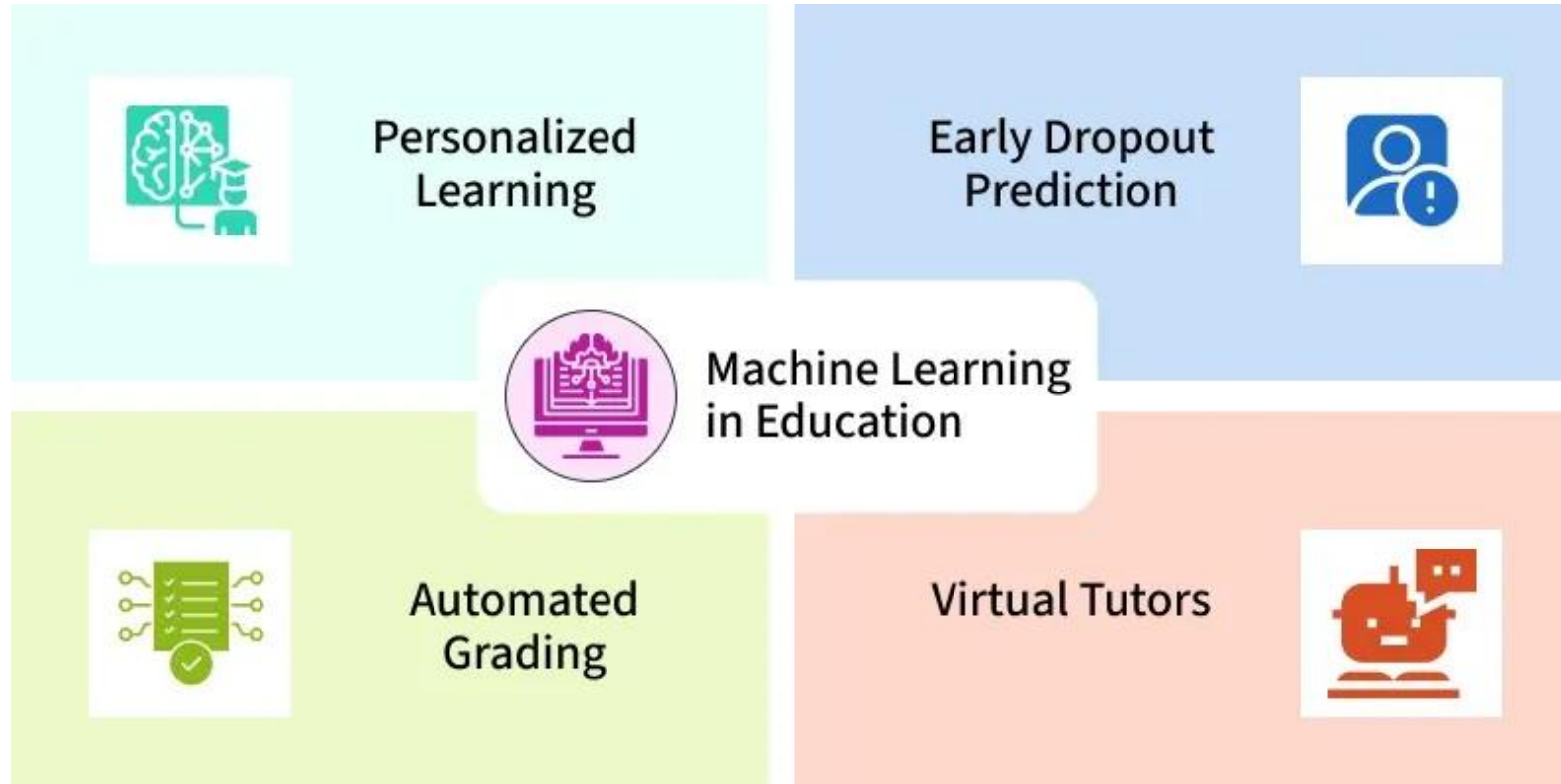


Drug Discovery



Virtual Assistants &
Chatbots

Applications



Applications



Perception of Surroundings



Object Detection & Classification



Machine Learning in Autonomous Vehicles



Navigation & Path Planning



Adapting to Different Conditions

Applications



Targeted
Advertising



Content
Recommendation



Machine Learning
in Marketing &
Advertising



Dynamic
Pricing



Chatbots &
Assistants